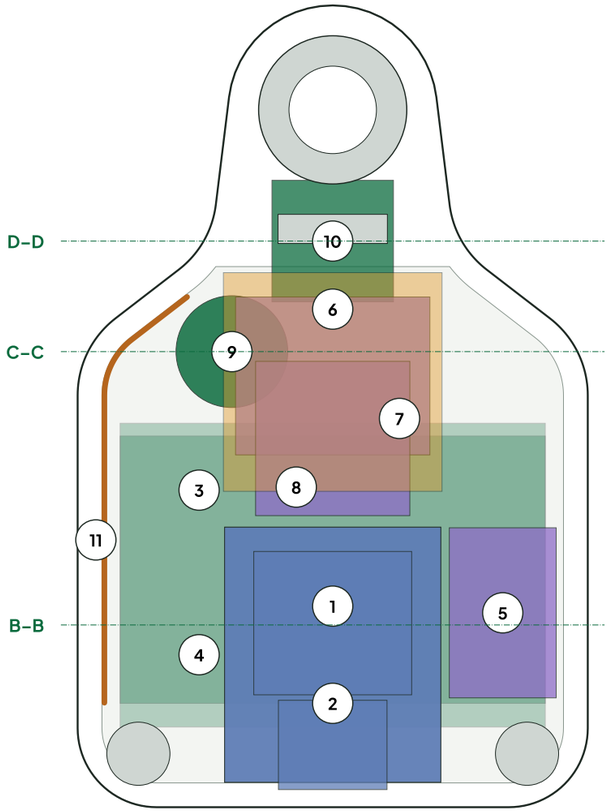


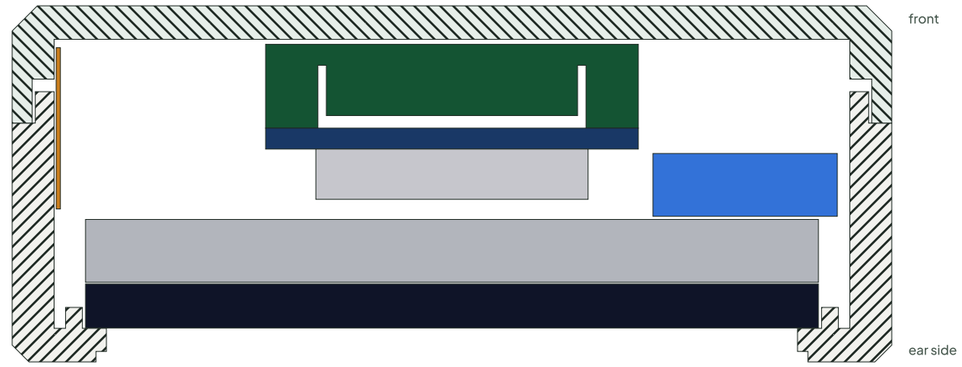
INTERNAL LAYOUT — FRONT VIEW, COVER REMOVED

scale 2.3 : 1

rear layer (ear / solar side) · MCU + radio stack · middle layer · front layer (behind logo) · side wall

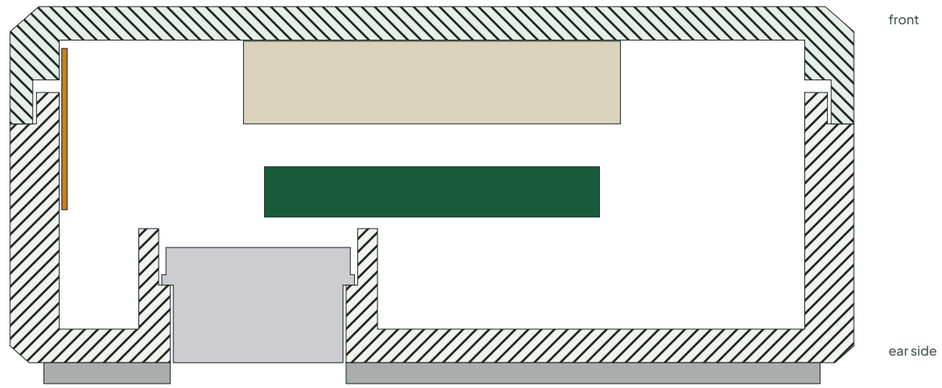
#	Component	Envelope (mm)	Function
1	Seeed XIAO ESP32S3	21.0 × 17.8 × 4.3	MCU, Li-ion charger, USB-C
2	Seeed Wio-SX1262 (XIAO kit)	21.0 × 17.8 × 4.0	LoRa / LoRaWAN AS923, u.FL
3	LiPo 302535 + PCM	35 × 25 × 3.0	3.7 V ~250 mAh, JST-SH
4	Solar cell ANYSOLAR SM141K05L	35 × 22 × 2.1	154 mW, Voc 3.4 V
5	Solar harvester (BQ25570 class)	≤ 14 × 9 × 3	boost MPPT charger
6	GNSS ceramic patch	18 × 18 × 4.0	passive, u.FL, faces out
7	GNSS module L76K / ATGM336H	≤ 16 × 13 × 2.4	GPS + BeiDou, UART
8	6-axis IMU breakout	≤ 12.7 × 12.7 × 2.0	MPU-6050 class, I²C
9	MLX90614ESF (TO-39)	Ø9.2 × 5.5	IR skin temp, I²C
10	MAX30102 mini module	≤ 10 × 10 × 3.2	HR + SpO_2 (PPG), I²C
11	LoRa FPC antenna 915/923	≤ 40 × 8 × 0.2	bent along the left wall

B–B



SECTION B–B · MCU, battery, solar · Y = 15

C–C



SECTION C–C · IR sensor + GNSS tower · Y = 37.5

D–D



SECTION D–D · PPG window + cover hook · Y = 46.6

Terrava

SMART EAR TAG · TV-ET0 CLASSIC

Internal layout & cross-sections

Every part checked for zero interference in CAD · Internal volume 21 cm³

UNITS	SCALE	TOLERANCE	REV	DATE	SHEET
mm	2.3 : 1 / 2.5 : 1	±0.2 unless noted	A	2026-10-21	2 / 3